

Arjuna JEE 2025

Maths

Sequence & Series

DPP: 3

Q1 The 6^{th} term of a GP is 32 and its 8^{th} term is 128, then the common ratio of the GP is

- (A) -1 (B) 2
(C) 4 (D) -4

Q2 If fifth term of a GP is 2, then the product of its 9 term is

- (A) 256 (B) 512
(C) 1024 (D) None of these

Q3 If a, b and c are in GP, then

- (A) a^2, b^2, c^2 are in GP
(B) $a^2(b+c), b^2(c+a), c^2(a+b)$ are in GP
(C) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in GP
(D) None of these

Q4 Direction $1, \log_y x, \log_z y, -15 \log_x z$ be the first four terms of an AP with common difference d . All the terms of an AP being real. Common difference d of an AP satisfies the equation

- (A) $6d^3 + 11d^2 + 6d - 16 = 0$
(B) $6d^3 + 11d^2 - 6d - 16 = 0$
(C) $6d^3 + 11d^2 + 6d + 16 = 0$
(D) $6d^3 + 11d^2 - 6d + 16 = 0$

Q5 Direction :

$1, \log_y x, \log_z y, -15 \log_x z$ be the first four terms of an AP with common difference d . All the terms of an AP being real. The value of $(x^2 y^2)$ is

- (A) 0 (B) 1
(C) 2 (D) 3

Q6

The number which should be added to the numbers 2, 14, 62 so that resulting numbers are in GP is

- (A) 1 (B) 2
(C) 3 (D) 4

Q7 If a, b, c are three unequal numbers such that a, b, c are in AP and $b-a, c-b, a$ are in GP then $a : b : c =$

- (A) $1 : 2 : 3$
(B) $1 : 3 : 4$
(C) $2 : 3 : 4$
(D) $1 : 2 : 4$

Q8 The n^{th} terms of a GP is 128 and the sum of its n terms is 255. If its common ratio is 2 then find the first term.

- (A) 1 (B) 2
(C) 3 (D) 4

Q9 The third term of a G. P. is 3. The product of its first five terms is

- (A) 81 (B) 243
(C) 729 (D) 343

Q10 If the second and fifth terms of a GP are 24 and 3 respectively, then the sum of first six terms is

- (A) 181
(B) $\frac{181}{2}$
(C) 189
(D) $\frac{189}{2}$

Q11 The sum $1 + 3 + 3^2 + \dots + 3^n$ is equal to

- (A) 3^{n+1}
(B) $\frac{3^{n+1}-1}{2}$



- (C) $3^{n+1} - 1$
 (D) 3°

Q12 If the sixth term of a GP be 2, then the product of first eleven terms is

- (A) 1024 (B) 2047
 (C) 2048 (D) 1023

Q13 How many terms of the series : $\sqrt{3}, 3, 3\sqrt{3}, \dots$ amounts to $39 + 13\sqrt{3}$?

- (A) 8 (B) 9
 (C) 6 (D) 10

Q14 The fifth terms of a GP. is 32 and common ratio is 2, then the sum of first 14 terms of the G. P. is

- (A) 16388 (B) 32667
 (C) 32766 (D) 64342

Q15 If the sum of first three numbers in GP is 21 and their product is 216, then the numbers are

- (A) 3, 6, 12 (B) 5, 7, 9
 (C) 6, 2, 213 (D) 6, 12, 24

Q16 If a, b and c are in G. P., then $\frac{b-a}{b-c} + \frac{b+a}{b+c} =$

- (A) $b^2 - c^2$
 (B) ac
 (C) ab
 (D) 0

Q17 If the continued product of three number in G. P. is 216 and the sum of the product taken in pairs is 156, then find the numbers.

- (A) 2, 6, 18
 (B) 18, 6, 4
 (C) 3, 8, 9
 (D) None of these

Q18 If the first term of a GP is 5 and common ratio is -5 , then which of the following term is 3125?

- (A) 6^{th}
 (B) 5^{th}

- (C) 7^{th}
 (D) 8^{th}

Q19 Three numbers are in GP such that their sum is 38 and their product is 1728. The greatest number among them is

- (A) 18 (B) 16
 (C) 14 (D) None of these

Q20 $0.2 + 0.22 + 0.222 + \dots$ upto n terms is equal to

- (A) $\left(\frac{2}{9}\right) - \left(\frac{2}{81}\right)(1 - 10^{-n})$
 (B) $n\left(\frac{1}{9}\right)(1 - 10^{-n})$
 (C) $\left(\frac{2}{9}\right)\left[n - \left(\frac{1}{9}\right)(1 - 10^{-n})\right]$
 (D) $\left(\frac{2}{9}\right)$



Answer Key

Q1 (B)

Q2 (B)

Q3 (A)

Q4 (C)

Q5 (B)

Q6 (B)

Q7 (A)

Q8 (A)

Q9 (B)

Q10 (D)

Q11 (B)

Q12 (C)

Q13 (C)

Q14 (C)

Q15 (A)

Q16 (D)

Q17 (A)

Q18 (B)

Q19 (A)

Q20 (C)



[Android App](#)

| [iOS App](#)

| [PW Website](#)